

Qatar-4b

R-2702

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_230812 · created 2026-09-01T00:24:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460168.9046 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.054
Transit depth (Rp/Rs)^2	3.21 +/- 1.92 [%]
Orbital Inclination (inc)	84.81 +/- 2.16
Airmass coefficient 1 (a1)	1.51 +/- 0.21
Airmass coefficient 2 (a2)	-0.42 +/- 1.06
Scatter in the residuals of the lightcurve fit is	14.14 %
Best Comparison Star	#3 - [210, 391]
Optimal Aperture	1.54
Optimal Annulus	6.18
Transit Duration (day)	0.076 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.054 vs 0.138 (deviation 0.76σ)
Duration fit vs published	0.076 d vs 0.0892 d (deviation 0.73σ)
Mid-time O–C	13.1 min
Depth SNR	3.3
Residual scatter	14.14 %

Light curve

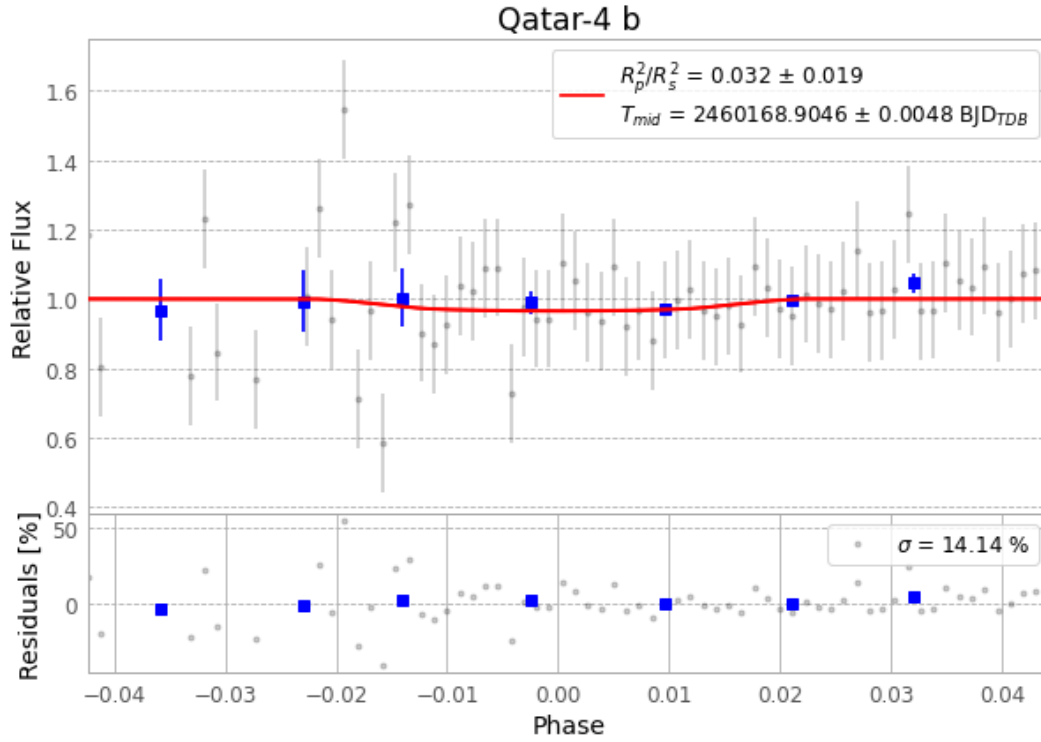


Figure 1 — FinalLightCurve_Qatar-4 b_2023-08-12.png (SHA-256 9aaca30a6653de19...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [244, 222], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 21e71721cb7afd82...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

76 FITS frames (50 MB), Qatar-4230812074816.FITS → Qatar-4230812113315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-230812.log (f84881f55d09...), FinalLightCurve_Qatar-4 b_2023-08-12.png (9aaca30a6653...), FinalLightCurve_Qatar-4 b_2023-08-12.csv (8f81b7c23b53...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:24Z.